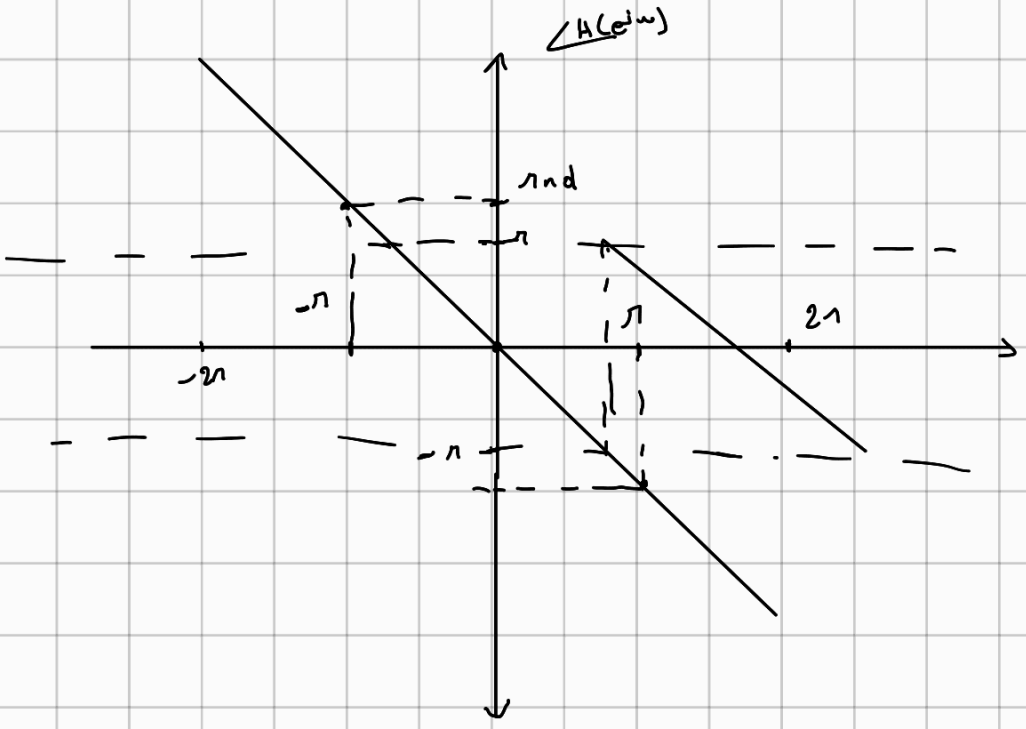
Örnek

$h_{id} = \delta[n-n_d]$

$H_{id}(e^{jw}) = e^{-jwn_d}$

$|H(e^{jw})| = 1$

$\angle H(e^{jw}) = -w n_d$

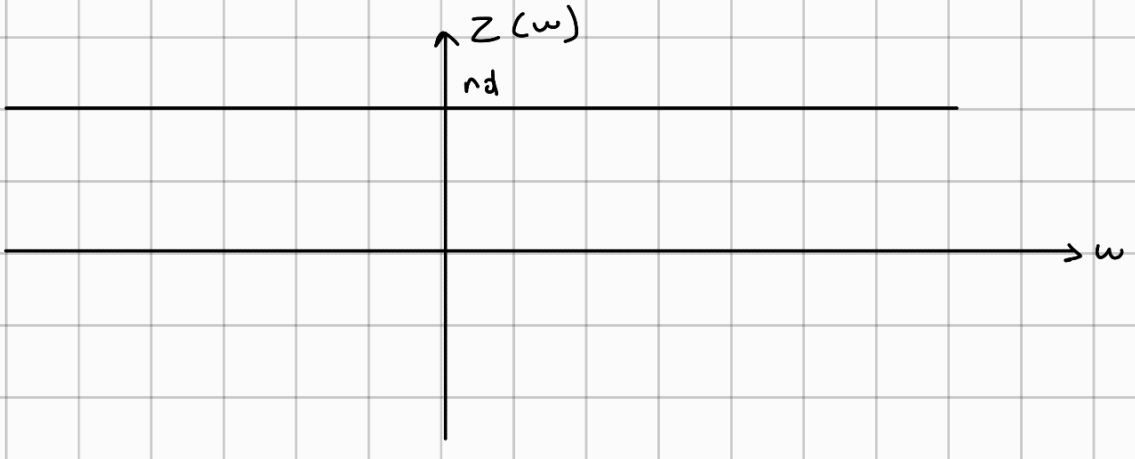


Groupdelay $\tau(w)$

$\tau(w) \triangleq \frac{-d}{dw} \angle H(e^{jw})$

For ideal delay system

$\tau(w) = \frac{-d}{dw}(-w n_d) = n_d$ [sample]



$h_{id}[n] = \delta[n-n_d]$

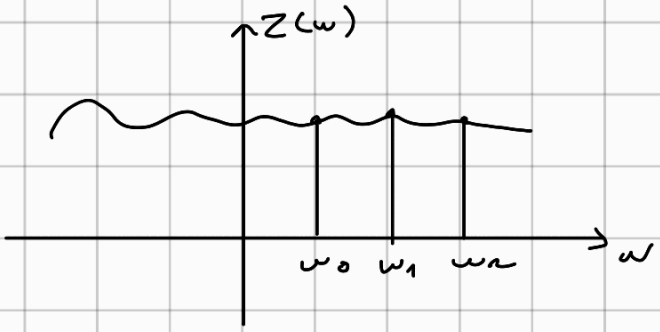
$x[n] = \cos(w_0 n)$

$H(e^{jw}) = e^{-jwn_d}$

↳ $g[n] = \cos(\omega_0 n - \omega n_d) = \cos(\omega_0 (n - n_d))$

$x[n] = \cos(\omega_0 n) + \cos(\omega_1 n) + \cos(\omega_2 n)$

$g[n] = \cos(\omega_0 (n-n_d)) + \cos(\omega_1 (n-n_d)) + \cos(\omega_2 (n-n_d))$



IIR hiç bir zaman linear faza sahip değildir

FIR sahip olabilmektedir

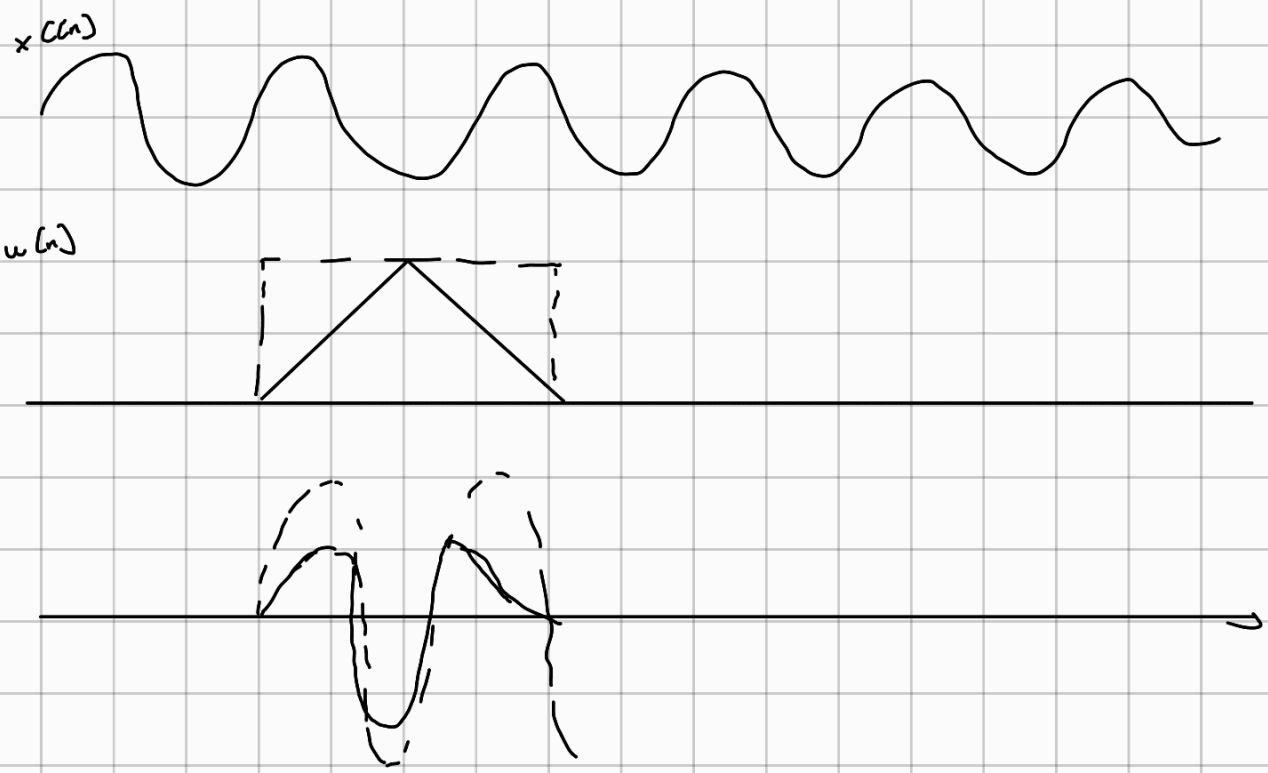
$\angle H(e^{jw}) = -\alpha w$

$\angle H(e^{jw}) = -\alpha w + \beta$

$x_1[n] = w[n] \cos(0,2\pi n)$

$x_2[n] = w[n] \cos(0,4\pi n - \pi/2)$

$x_3[n] = w[n] \cos(0,8\pi n + \pi/5)$

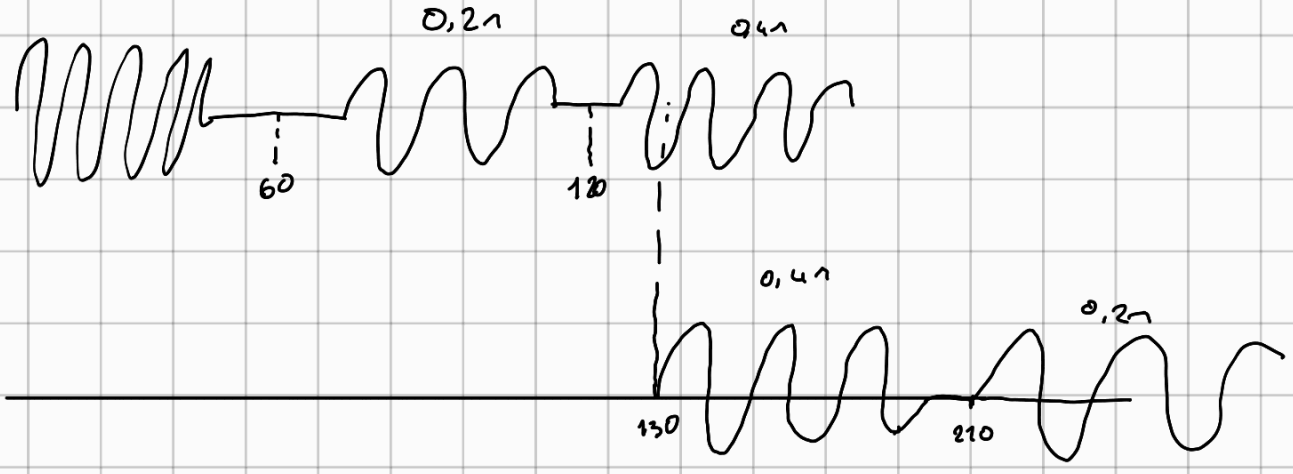


$$x[n] = x_3[n] + x_1[n-M-1] + x_2[n-2M-2]$$

$M = 60$

ya var bağımlılık kuralı

$$x[n] = x_3[n] + x_1[n-60] + x_2[n-120]$$





$0,4\pi$ 'de $n_d = 10$, $\angle H(e^{j0,4})$

$= -0,4\pi \times 10 = -4\pi$

Frequency Response

$$\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{k=0}^{M} b_k\, x[n-k]$$

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{\sum_{k=0}^{N} a_k e^{-j\omega k}} = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M} (1 - c_k e^{-j\omega})}{\prod_{k=1}^{N} (1 - d_k e^{-j\omega})}$$

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

$$20 \log |H(e^{jw})| = 10 \log |H(e^{jw})|^2$$

$$|H(e^{jw})|^2 = H(e^{jw}) H^*(e^{jw})$$

$|H(e^{jw})| = 1 \Rightarrow 0\,db$

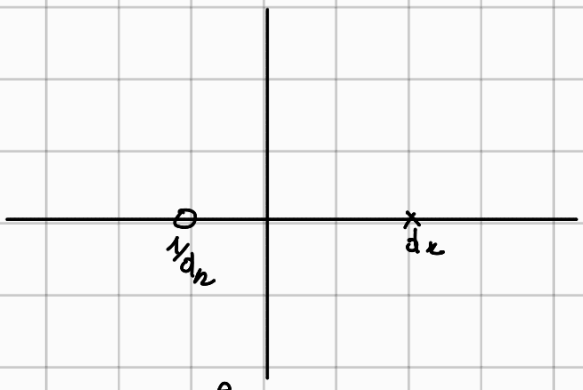
$|H(e^{jw})| = \frac{1}{\sqrt{2}} \Rightarrow -3\,db$

$$20 \log |H(e^{jw})| = 20 \log \left(\frac{b_0}{a_0}\right) + \sum_{k=1}^{M} 20 \log |1 - c_k e^{-jw}| - \sum_{k=1}^{N} 20 \log |1 - d_k e^{-jw}|$$

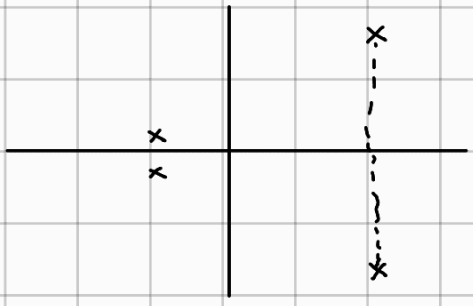
$d_k = r e^{j\theta}$

$$|1 - r e^{j\theta} e^{-jw}|^2 = (1 - r e^{j\theta} e^{-jw})(1 - r e^{-j\theta} e^{jw})$$

$$= 1 + r^2 - 2r \cos(w - \theta)$$



$$H(z) = \prod_{k=1}^{p} \frac{z^{-1} - d_k}{1 - d_k z^{-1}}$$



$$H(z) = \frac{(z^{-1} - e_k^*)(z^{-1} - e_k)}{(1 - e_k z^{-1})(1 - e_k^* z^{-1})}$$

$|H(e^{jw})| = 1$

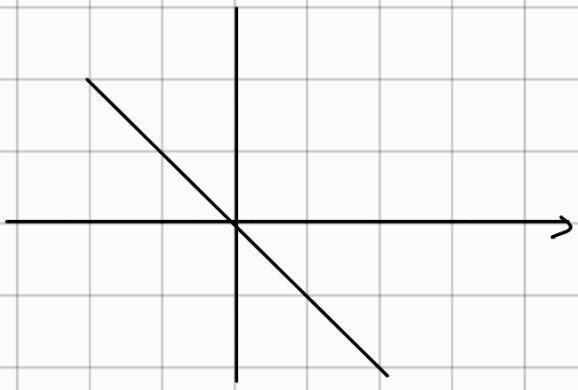
$$H(e^{jw}) = \frac{e^{-jw} - d_k}{1 - d_k e^{-jw}}$$

$$|H(e^{jw})|^2 = \frac{e^{-jw} - d_k}{1 - d_k e^{-jw}} \cdot \frac{e^{jw} - d_k}{1 - d_k e^{jw}}$$

ALLpass

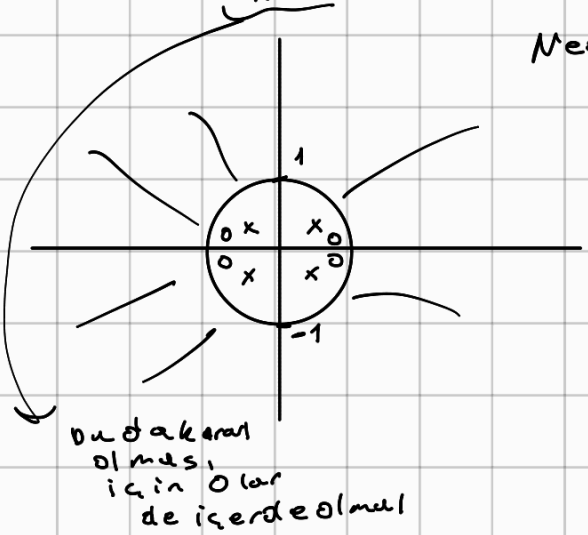


Ama çözü farklı olabili.



Minimum phase systems

$H(z) \rightarrow \frac{1}{H(z)}$



Nedenci ve kararlı

Bu da kararlı olması için 0 lar da içerde olmalı

Hem kutuplar hem 0'ların birim çemberi içinde olması şart

Linear Phase filters

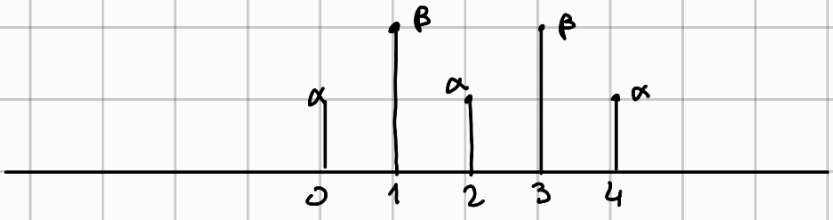
Type-I

$h[n] = h[M-n] \qquad 0 \le n \le M \qquad M$ even integer

$M = 4 \Rightarrow 0 \le n \le 4$

$h[0] = h[4]$

$h[1] = h[3]$



Type II

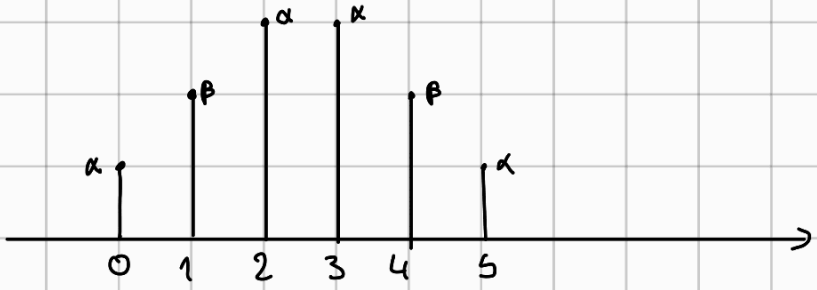
$h[n] = h[M-n]$ $\quad 0 \le n \le M$ → odd integer

$M = 5$

$h[0] = h[5]$

$h[1] = h[4]$

$h[2] = h[3]$



Type III

$h[n] = -h[M-n]$

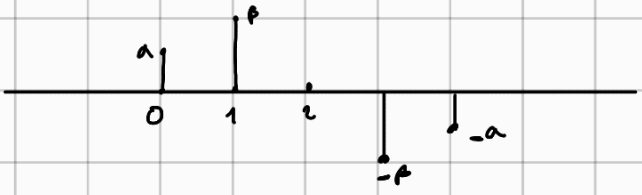
M even odd

$M = 4$

$h[0] = -h[4]$

$h[1] = -h[3]$

$h[2] = 0$



Type IV

$h[n] = -h[M-n]$

M odd integer

(tip 2)

Show $h[n] = h[M-n]$, M odd

→ get linear phase

$M=5$ $\quad h[0] = h[5]$

$h[1] = h[4]$

$h[2] = h[3]$

$$H(e^{j\omega}) = \sum_{n=0}^{M} h[n] e^{-j\omega n} = \sum_{n=0}^{\frac{M-1}{2}} h[n] e^{-j\omega n} + \sum_{n=\frac{M+1}{2}}^{M} h[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\frac{M-1}{2}} h[n] e^{-j\omega n} + \sum_{k=0}^{M-\frac{1}{2}} h[M-k] e^{-j\omega(M-k)}$$

$$= \sum_{n=0}^{\frac{M-1}{2}} h[n] e^{-j\omega n} + \left(\sum_{n=\infty}^{\frac{M-1}{2}} h[n] e^{+j\omega n} \right) e^{-j\omega M}$$

$$= \sum_{n=0}^{\frac{M-1}{2}} h[n] \left[e^{-j\omega n} + e^{j\omega n} \; e^{-j\omega M} \right]$$

$$= \sum_{n=0}^{\frac{M-1}{2}} h[n] \left[e^{-j\omega n} \; e^{-j\omega \frac{M}{2}} \; e^{j\omega \frac{M}{2}} + e^{j\omega n} \; e^{-j\omega \frac{M}{2}} \; e^{-j\omega \frac{M}{2}} \right]$$

$$= e^{-j\omega \frac{M}{2}} \sum_{n=0}^{\frac{M-1}{2}} h[n] \left[e^{-j\omega(n-\frac{M}{2})} + e^{j\omega(n-\frac{M}{2})} \right]$$

$$H(e^{j\omega}) = e^{-j\omega \frac{M}{2}} \; 2 \sum_{n=0}^{\frac{M-1}{2}} h[n] \cos\left(\omega n - \omega \frac{M}{2}\right)$$

$$|H(e^{j\omega})| = \left| 2 \sum_{n=0}^{\frac{M-1}{2}} h[n] \cos\left(\omega n - \omega \frac{M}{2}\right) \right|$$

$$\angle H(e^{j\omega}) = \begin{cases} -\omega \frac{M}{2} & > 0 \\ -\omega \frac{M}{2} + \pi & < 0 \end{cases}$$